NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Cloud Computing

Assignment- Week3

TYPE OF QUESTION: MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1:

Which of the following is false about SLA contents

a) A complete, specific definition of each service
b) An auditing mechanism to monitor the services
c) The responsibilities of the service provider and consumer
d) None of these

Correct Answer: d
Detailed Solution: All of these are part of SLA

QUESTION 2:

In cloud, computing resources are allocated

a) Through a centralized service discovering agent
b) By service level agreement
c) Globally without any centralized agent
d) All of them

Correct Answer: c
Detailed Solution: Cloud related resources are allocated and distributed globally without any central directory or agent

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QUESTION 3:

Which of the following is correct about Availability?

a) Availability = 1- (UpTime / TotalTime)
b) Availability = 1+ (DownTime / UpTime)
c) Availability = 1+ (UpTime / DownTime)
d) Availability = 1- (DownTime / UpTime)

Correct Answer: d

Detailed Solution: Availability = 1- (DownTime / UpTime). Correct option is (d).

QUESTION 4:

Statement 1: Coefficient of variance represents the ratio of standard deviation to mean.
Statement 2: Coefficient of variance is a useful statistic for comparing the degree of uniformity from one data series to another.

a) Only statement 1 is true
b) Only statement 2 is true
c) Both the statements are true
d) Both the statements are False

Correct Answer: a

Detailed Solution: Coefficient of variance is a useful statistic for comparing the degree of variation from one data series to another. Hence, the correct option is (a).

QUESTION 5:

Say a cloud service is well maintaining it's 95% availability as mentioned in Service Level Agreement. However, still the services are not performing as expected when the load increases abruptly. The problem can be referred to as an issue of

a) Load balancing
b) Linearity
c) Reliability
d) Durability

Correct Answer: b

Detailed Solution: Linearity ensures how system performs when load increases.

QUESTION 6:

Which of the following is true about penalty, where D(t) is instantaneous demand at time t and R(t) is response at time t?

a) Penalty $\propto \int | R(t) * D(t) | dt$
b) Penalty $\propto \int | D(t) + R(t) | dt$
c) Penalty $\propto \int | D(t) – R(t) | dt$

d) None of these

Correct Answer: c

Detailed Solution: Penalty is proportional to ∫ | D(t) – R(t) | dt

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QUESTION 7:

It is absolutely beneficial to purchase a cloud service when

a) Utility premium < the peak to average demand ratio
b) Utility premium > the peak to average demand ratio
c) Utility premium <= the peak to average demand ratio
d) Utility premium >= the peak to average demand ratio

Correct Answer: a

Detailed Solution: It is always better to purchase cloud service when the utility premium is less than the peak demand to average demand ratio. In case they are equal, we can go for either cloud or self-owned system.

QUESTION 8:

If R_t is the ratio of execution time taken by a process in sequential and parallel architecture, and p is the number of parallel computing nodes available, then the parallel efficiency is

a) R_t / p
b) p / R_t
c) None of the above
d) $p \times R_t$

Correct answer: a

Detailed Solution: Parallel efficiency is (time taken in for sequential execution) / (no of processor x time taken in parallel execution)

QUESTION 9:

Statement 1: Hadoop Distributed File System is an open source implementation of GFS architecture.

Statement 2: In Google File System, the master maintains regular communication with the chunk servers.

a) Only statement 1 is true
b) Only statement 2 is true
c) Both the statements are true
d) Both the statements are false

Correct Answer: c

Detailed Solution: Hadoop Distributed File System is an open source implementation of GFS architecture. In the Google File System, the master maintains regular communication with the chunk server. This is better known as heartbeat messages. In case of a failure, chunk server metadata gets updated to reflect failure.

QUESTION 10:

Which of the following is true?

a) IaaS can be built on PaaS
b) PaaS can be built on SaaS
c) IaaS can be built on SaaS
d) None of the above

Correct Answer: d

Detailed Solution: PaaS can be built on IaaS, and SaaS can be built on PaaS.

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